

START — USER OPENS AI-IGMS

Login Page — Enter Email, Password, Select Role

Send POST /api/auth/login (email, password)

Express Server Receives → CORS → JSON Parser → Rate Limiter → Morgan
Logger

Router matches /api/auth/login → authController.login()

Find user: User.findOne({email}).select('+password')

User exists?

NO → 401: Invalid email or password

↓ YES

Account is Active?

NO → 403: Account deactivated

↓ YES

bcrypt compare
matches?

NO → 401: Invalid email or password

↓ YES

Generate JWT: jwt.sign({id}, SECRET, {expiresIn: 7d})

Response: {success: true, token, user} — password stripped by toJSON()

Browser stores JWT in localStorage → AuthContext
updates → ProtectedRoute reads role

User Role?

SUPER ADMIN

Manage Schools (CRUD)

GET/POST /api/schools schools

Assign Principals

PATCH /api/schools users

View Network Reports

GET /api/reports users, schools

Register Users

POST /api/auth/register users

PRINCIPAL

Manage Students (CRUD)

/api/students users

Manage Teachers (CRUD)

/api/teachers users

Post Notices

POST /api/notices notices

Approve Leaves

PATCH /api/leave/:id leave

View Reports

/api/reports all

TEACHER

Mark Attendance

POST /api/attendance attendances

Upload Marks

POST /api/marks marks

Create Online Exam

POST /api/exams exams

Evaluate Submissions

GET /api/submissions submissions

Apply Leave

POST /api/leave leaves

STUDENT

View My Attendance

GET /api/attendance/me attendances

View My Results

GET /api/results/me marks

Attempt Online Exam

POST /api/exams/:id/submit submissions

View Notices

GET /api/notices notices

PARENT

View Child Overview

GET /api/users/child users

View Child Attendance

GET /api/attendance/child attendances

View Child Results

GET /api/results/child marks

View Notices

GET /api/notices notices

ERROR HANDLING PIPELINE

Error Thrown

asyncHandler catches

errorHandler middleware

JSON Response
(400/401/403/404/500)

Frontend: toast.error() — page
stays, nothing crashes

All Requests: Bearer JWT in header → protect middleware (jwt.verify) → authorize(role)

Controller processes → Mongoose ODM query

